

XTSF: Synthetic Forecasting Contributions with XPC Theory, Estimands, Implementation, and Experiments

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Abstract

XTSF studies whether Monte Carlo Shapley explanations recover known contributions in a controlled forecasting problem. A synthetic hourly load is generated as the sum of explicit weather, calendar, trend, and irrelevant components, so the structural contribution of every group is known. The experiments separate two sources of error: Monte Carlo error relative to the chosen Shapley estimand, and estimand mismatch relative to the structural decomposition. They compare interventional and nearest-neighbor conditional Shapley values, vary the number of sampled coalitions and masking samples, and reproduce three aggregation games from the original XPC work. The accompanying `xpc` package provides the reusable data, model, masking, grouping, estimation, and post-processing abstractions used by the notebook.

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1 Research Question and Scope

The main question is not merely whether a Shapley estimator converges. It is whether the quantity to which it converges corresponds to the contribution of interest. For an observation x , a forecasting model f , and a known structural decomposition $C^{\text{struct}}(x)$, the project asks:

1. How quickly does Monte Carlo Shapley converge as the number of coalitions and masking samples increases?
2. How does an interventional coalition value differ from a conditional coalition value when features are dependent?
3. How close is each exact Shapley estimand to the known structural contribution?
4. How do alternative aggregation games trade computation for a changed estimand?

The word *contribution* therefore refers to three distinct objects:

- the structural contribution specified by the synthetic data-generating process;
- the exact Shapley contribution for a chosen empirical coalition-value function;
- its finite-budget Monte Carlo estimate.

Keeping these objects separate is essential. A low error relative to an exact conditional Shapley value only establishes numerical convergence to that conditional estimand. It does not establish agreement with the structural or causal decomposition.

2 Theoretical Framework

2.1 Shapley Values

Let $N = \{1, \dots, p\}$ be a set of players and let $v_x(S)$ be the value of coalition $S \subseteq N$ for the observation x . The Shapley value of player j is

$$\phi_j(x; v) = \sum_{S \subseteq N \setminus \{j\}} \frac{1}{p \binom{p-1}{|S|}} [v_x(S \cup \{j\}) - v_x(S)]. \quad (1)$$

The weights in Equation (1) are equivalent to first drawing the coalition size uniformly from $0, \dots, p-1$, then drawing a subset of that size uniformly from the remaining players. For a single coherent game, exact Shapley values satisfy efficiency:

$$\sum_{j=1}^p \phi_j(x; v) = v_x(N) - v_x(\emptyset). \quad (2)$$

In the core package, players may be raw features or explicit disjoint feature groups. Changing the groups changes the cooperative game; groups are not merely display labels.

2.2 Baseline, Interventional, and Conditional Values

For a deterministic reference vector b , baseline masking defines

$$v_x^{\text{base}}(S) = f(x_S, b_{\bar{S}}). \quad (3)$$

For a random feature vector X , the interventional value is

$$v_x^{\text{int}}(S) = \mathbb{E}[f(x_S, X_{\bar{S}})]. \quad (4)$$

It preserves the dependence among jointly missing features, but breaks their dependence with the features fixed to x_S . Consequently, it may evaluate combinations that are uncommon in the observed data.

The conditional value is

$$v_x^{\text{cond}}(S) = \mathbb{E}[f(x_S, X_{\bar{S}}) \mid X_S = x_S]. \quad (5)$$

Conditional masking stays closer to the observed support because missing features are drawn from rows compatible with the present coalition. This can be more realistic for prediction explanations, but it is still associational. Observed dependence may arise from confounding, common causes, time, or feature construction. Conditional Shapley is therefore not automatically a causal attribution.

2.3 Exact Empirical Estimands

Let $B = \{z^{(1)}, \dots, z^{(R)}\}$ be the finite background data. The exact empirical interventional value used in the notebook is

$$\hat{v}_x^{\text{int}}(S) = \frac{1}{R} \sum_{r=1}^R f(x_S, z_{\bar{S}}^{(r)}). \quad (6)$$

For conditional values, the observed columns S are standardized by their background standard deviations. Let $\mathcal{N}_K(x_S)$ be the $K = 64$ nearest background rows in that standardized space. The notebook defines

$$\hat{v}_x^{\text{cond}}(S) = \frac{1}{K} \sum_{z \in \mathcal{N}_K(x_S)} f(x_S, z_{\bar{S}}), \quad (7)$$

with all background rows used when $S = \emptyset$. An exact empirical Shapley reference is obtained by evaluating these deterministic coalition values and enumerating every subset in Equation (1).

Here, *exact* means exact for the finite empirical or nearest-neighbor game. It removes coalition-sampling and mask-sampling noise. It is not an exact population conditional expectation and it is not a causal ground truth.

3 Synthetic Structural Ground Truth

3.1 Data-Generating Process

The notebook creates 120 days of hourly observations, for $R = 2880$ rows. Temperature combines a slow seasonal component, an hourly component, and Gaussian noise. This induces dependence

between weather variables, time of day, weekday status, and trend. The six model features and structural groups are summarized below.

Group	Feature	Definition or role	Coefficient
Weather	heating degree	$\max(18 - T, 0)$	1.20
Weather	cooling degree	$\max(T - 23, 0)$	1.80
Calendar	daily peak	morning and evening Gaussian basis functions	14.0
Calendar	weekday work	indicator for Monday through Friday	6.0
Trend	trend	linear ramp from 0 to 1	10.0
Irrelevant	irrelevant	independent standard Gaussian feature	0.0

Table 1: Synthetic forecasting features and structural coefficients.

The load and forecasting model are identical and contain no observation noise:

$$f(x) = 60 + 1.2x_{\text{heat}} + 1.8x_{\text{cool}} + 14x_{\text{peak}} + 6x_{\text{weekday}} + 10x_{\text{trend}}. \quad (8)$$

The true group contributions are the corresponding additive terms:

$$C_{\text{weather}}(x) = 1.2x_{\text{heat}} + 1.8x_{\text{cool}}, \quad (9)$$

$$C_{\text{calendar}}(x) = 14x_{\text{peak}} + 6x_{\text{weekday}}, \quad (10)$$

$$C_{\text{trend}}(x) = 10x_{\text{trend}}, \quad (11)$$

$$C_{\text{irrelevant}}(x) = 0. \quad (12)$$

Thus the structural decomposition is known for every row. With zero-baseline masking, the grouped Shapley values equal these terms exactly because the model is additive and every basis feature is absent at zero. This is used as an oracle sanity check of tensor shapes and grouping behavior.

3.2 Putting Signed Shapley Values on the Structural Scale

Empirical Shapley values explain deviations from the empty-coalition value. The notebook compares them with positive structural contributions by restoring the matching background mean:

$$\hat{C}_g(x) = \hat{\phi}_g(x) + \bar{C}_g, \quad \bar{C}_g = \frac{1}{R} \sum_{r=1}^R C_g(z^{(r)}). \quad (13)$$

For the additive model, exact interventional Shapley satisfies $\phi_g^{\text{int}}(x) = C_g(x) - \bar{C}_g$, so Equation (13) recovers structural truth. Conditional Shapley may reallocate signal across dependent groups, so the same mean allocation does not generally recover each structural component.

Two mean squared errors are reported over evaluation rows i and groups g :

$$\text{MSE}_{\text{struct}} = \frac{1}{nG} \sum_{i,g} \left(\hat{C}_{i,g} - C_{i,g}^{\text{struct}} \right)^2, \quad (14)$$

$$\text{MSE}_{\text{estimand}} = \frac{1}{nG} \sum_{i,g} \left(\hat{C}_{i,g} - C_{i,g}^{\text{exact}} \right)^2. \quad (15)$$

The second quantity measures Monte Carlo error. The gap between the exact estimand and structural truth is an estimand mismatch that more computation cannot remove.

4 Monte Carlo Implementation

4.1 Estimator

For every explained unit and player g , the estimator draws $K = \text{n_coalitions}$ coalitions from the Shapley subset distribution. For each coalition it estimates

$$\hat{v}_x(S) = \frac{1}{M} \sum_{m=1}^M f(\tilde{x}_S^{(m)}), \quad M = \text{n_mask_samples}, \quad (16)$$

then averages the marginal differences

$$\hat{\phi}_g(x) = \frac{1}{K} \sum_{k=1}^K [\hat{v}_x(S_k \cup \{g\}) - \hat{v}_x(S_k)]. \quad (17)$$

Increasing `n_coalitions` reduces subset-sampling noise. Increasing `n_mask_samples` reduces noise in the empirical coalition-value expectation. The implementation samples the before and after coalition values independently.

4.2 Package Abstractions

The installable package is named `xpc`. Its main responsibilities are:

Abstraction	Implemented responsibility
<code>DataSpec</code>	Tabular (n, f) data and normalized (n, H) outputs.
<code>TimeSeriesTensorSpec</code>	Temporal (n, d, f) data; masks one time row while the model receives the full sequence, returning (n, d, H, G) explanations.
<code>ModelAdapter</code>	NumPy boundary for callables, pandas, sklearn-style, PyTorch, external-script, R-script, and registered custom models.
<code>Masker</code>	Deterministic baseline, unconditional empirical, or conditional empirical replacement.
<code>Conditioner</code>	Nearest-neighbor, grid-cell, or user-defined non-generative conditional selection.
<code>FeatureGroups</code>	Disjoint named players, remaining-feature policies, and always-present features.
<code>ShapleyExplainer</code>	Monte Carlo subset estimator returning signed values and optional heightened parts.
<code>Explanation</code>	Values, base values, predictions, efficiency residuals, heightening, and comparison metrics.

Table 2: Current XPC implementation boundaries.

For temporal inputs, background rows used by empirical maskers are drawn from the flattened collection of time locations. Conditional matching operates on the features of the currently explained row; it is not a temporal generative model.

Always-present features are excluded from the player set and restored in every coalition. They receive no separate attribution. They affect the base value and, in models with interactions, may also change the marginal contributions assigned to other players.

4.3 Explanation and Efficiency Diagnostics

An `Explanation` stores signed values Φ , base values b , and model predictions $\hat{y}$. Its efficiency residual is $r = \hat{y} - (b + \sum_g \hat{\phi}_g)$.

For a coherent exact Shapley game this residual is zero by Equation (2). In the package it diagnoses finite Monte Carlo error, independently estimated coalition values, and model-output shape handling. Changing from interventional to conditional masking changes the estimand, but does not by itself invalidate Shapley efficiency.

4.4 Heightening

Heightening is optional positive post-processing and is enabled by default. For every output and group, the implementation computes an offset over all batch axes,

$$o_{h,g} = \alpha \max\left(-\min_u \Phi_{u,h,g}, 0\right), \quad \alpha \geq 0, \quad (18)$$

then forms non-negative values, percentages, and target allocations:

$$\Phi_{u,h,g}^+ = \max(\Phi_{u,h,g} + o_{h,g}, 0), \quad (19)$$

$$p_{u,h,g} = \frac{\Phi_{u,h,g}^+}{\sum_j \Phi_{u,h,j}^+}, \quad (20)$$

$$\text{part}_{u,h,g} = p_{u,h,g} T_{u,h}. \quad (21)$$

The target defaults to the non-negative model prediction. If all positive values are zero, a uniform allocation is used. Heightening leaves the signed values untouched. All convergence and aggregation MSE experiments use signed values with `heighten=False`; heightened values are visualization outputs, not the estimand being benchmarked.

5 Convergence Experiment

The convergence benchmark uses the four structural groups in the direct grouped game. The full 2880-row panel is the background. Six evaluation rows are selected at evenly spaced indexes, and every setting is repeated five times with deterministic but distinct random seeds.

Two one-dimensional sweeps are performed:

- `n_coalitions` in $\{1, 2, 4, 8, 16, 32\}$, with `n_mask_samples=16`;
- `n_mask_samples` in $\{1, 2, 4, 8, 16, 32\}$, with `n_coalitions=16`.

For interventional Shapley, the additive structural contribution is also the exact empirical estimand after the mean allocation in Equation (13). For conditional Shapley, the notebook enumerates all coalitions and uses Equation (7) to construct a separate exact empirical reference.

As computation increases, the interventional structural MSE tends toward zero. Conditional MSE relative to its own estimand also decreases, but its structural MSE plateaus near the exact conditional estimand gap of 7.31. For example, at 32 coalitions and 16 mask samples, the conditional MSE is 0.172 relative to its estimand and 7.305 relative to structural truth. This is the central distinction of the experiment: convergence does not remove estimand bias.

6 Aggregation Games

The archived XPC work considered a climate group C and a non-climate group $D = N \setminus C$. The notebook reproduces three definitions.

6.1 Default Post-hoc Aggregation

First compute a Shapley value for every raw feature, then sum values belonging to the group:

$$\phi_C^{\text{post}}(x) = \sum_{j \in C} \phi_j(x; v). \quad (22)$$

This preserves the efficiency of the original raw-feature game after all groups are summed, but requires explaining every raw feature.

6.2 Coalitional Shapley

For each requested group C , merge all features in C into one player while every feature outside C remains an individual player. With $q_C = p - |C| + 1$,

$$\phi_C^{\text{coal}}(x) = \sum_{S \subseteq N \setminus C} \frac{1}{q_C \binom{q_C-1}{|S|}} [v_x(S \cup C) - v_x(S)]. \quad (23)$$

Climate and non-climate are computed in two different reduced games because their complements contain different raw players. Each value is a valid Shapley value in its own game, but the pair is not generally jointly efficient. This corrects the stronger efficiency claim made in the historical report.

6.3 Simplified Two-player Shapley

Treat C and D as the only two players in one common game:

$$\phi_C^{\text{two}}(x) = \frac{1}{2} ([v_x(C) - v_x(\emptyset)] + [v_x(N) - v_x(D)]), \quad (24)$$

$$\phi_D^{\text{two}}(x) = \frac{1}{2} ([v_x(D) - v_x(\emptyset)] + [v_x(N) - v_x(C)]). \quad (25)$$

Only the four coalition values $v(\emptyset), v(C), v(D), v(N)$ are needed. The pair is efficient because both values belong to the same two-player game.

The package’s direct grouped-player API generalizes this idea to any number of explicit groups through `ShapleyExplainer`. The exact historical post-hoc, coalitional, and two-player comparison is implemented locally in the notebook, not as an `aggregation` parameter in the core package.

7 Aggregation Experiment and Results

The fixed-budget comparison uses 16 sampled coalitions, 16 masking samples per coalition value, six evaluation rows, and five repetitions. The simplified mode does not sample coalitions because the two-player formula is exhaustive; its four coalition values are still estimated from 16 masked rows each.

Masking	Mode	Exact structural	MC structural	MC estimand	Rows/point
Interventional	Post-hoc	0.000	0.512	0.512	3072
Interventional	Coalitional	0.000	0.171	0.171	1024
Interventional	Two-player	0.000	1.594	1.594	64
Conditional	Post-hoc	11.398	12.478	1.362	3072
Conditional	Coalitional	10.943	10.718	0.573	1024
Conditional	Two-player	10.203	11.572	1.267	64

Table 3: Aggregation-mode mean squared errors. “Exact structural” is the exact empirical estimand’s MSE against structural truth. Monte Carlo columns are means over five repetitions.

The masked-row count follows directly from the implemented loops. Post-hoc uses $6 \times 16 \times 2 \times 16 = 3072$ model rows per point. Coalitional uses $2 \times 16 \times 2 \times 16 = 1024$. The two-player mode uses $4 \times 16 = 64$, a factor of 48 below post-hoc aggregation. Measured timings show the same ordering but are machine-dependent.

All interventional exact aggregation estimands coincide with structural truth because the model is additive. Conditional exact estimands differ because conditioning reallocates signal among dependent features and because the three aggregation definitions are different games. The smallest conditional exact structural MSE in this sample is not evidence that the corresponding mode is generally more causal. It is a property of this data, model, grouping, and nearest-neighbor estimand.

The exact conditional Coalitional pair has an efficiency RMSE of 0.842, while post-hoc and the common two-player game are efficient up to numerical precision. This confirms that independently computed Coalitional group values should not be interpreted as a single efficient decomposition.

8 Outputs and Reproducibility

The single Colab-ready notebook is `src/xtsf.ipynb`. It caches the synthetic panel and writes generated artifacts under `outputs/`:

- `outputs/synthetic_forecasting_panel.csv`;
- `outputs/shapley_convergence_runs.csv`;
- `outputs/shapley_convergence_summary.csv`;
- `outputs/shapley_convergence.png`;
- `outputs/aggregation_exact_estimands.csv`;
- `outputs/aggregation_mode_runs.csv`;
- `outputs/aggregation_mode_summary.csv`;
- `outputs/aggregation_modes.png`.

The package implementation is under `src/xpc/`; lightweight tests are under `src/tests/`. The public estimator deliberately excludes known contributions, SMACH-specific processing, GAM-specific paths, Kernel SHAP, Permutation SHAP, and generative conditional models. External or known contributions enter through `Explanation.from_contributions` and `Explanation.compare`, not through the estimator.

9 Interpretation and Limitations

- The exact structural decomposition is available because the synthetic model is additive and matches the data-generating process. More complex models with interactions will not make the three interventional aggregation estimands coincide.
- Conditional nearest-neighbor masking avoids many unsupported feature combinations, but remains a finite-sample associational approximation. It does not identify interventions or causal effects.
- Exact empirical references remove Monte Carlo noise but retain finite background, neighbor-selection, model, and grouping choices.
- The benchmarks use six evaluation points and five repetitions. They demonstrate estimand and computation effects, but are not a definitive large-sample performance study.
- Heightened positive parts are useful forecast allocations, but they are post-processed visual quantities. Scientific error comparisons use the signed Shapley values plus the documented background-mean allocation.